

An RGB image-based phenotyping framework for dry root rot disease assessment

Methods

Image acquisition

1. Carefully uproot the plants to avoid any damage to the root system.
2. Gently wash uprooted plants under running tap water to remove adhering soil particles without damaging the roots.
3. Capture images of cleaned root samples using three imaging modalities:
 - a. Handheld camera: Fujifilm X-T200
 - b. Root scanner: Epson Perfection V850 Pro
 - c. Microscope: Euromex DC 1355 F050

Image data curation

1. Remove corrupted or blurred images from the handheld camera dataset.
2. Crop retained images to eliminate background regions and non-root artifacts (e.g., soil debris or scale bars).
3. For images containing multiple roots, crop individual regions to generate images with varying numbers of root instances.
4. For root scanner images, remove shoot portions and retain only root tissue.
5. If multiple roots are present in a single scanner image, crop individual root instances separately.
6. Do not perform additional curation for microscopic images, as each image represents a single root system.

Disease scoring

Manual disease scoring

1. Observe the root necrosis symptoms.

2. Score for the disease manually by visual observation of the degree of necrosis on a 1-9 scale based on symptom intensity and extent of root damage (Table 1).

Disease scoring using WinRHIZO

1. Run the WinRHIZO Pro 2022 software on the computer.
2. Navigate to the color tab to create a new color analysis class (CAC) file.
3. Define color classes ranging from 0 to 9 corresponding to increasing levels of root necrosis (see **Note 3**).
4. Specify root tissue and background color ranges and create the CAC file.
5. Save the active set of color classes and groups to a file on disk which is saved with .cac extension for future reference and analysis.
6. Go to the color tab, select load classes and locate the CAC file created.
7. Navigate to the acquisition icon, locate and upload the image for analysis (see **Note 4**).
8. Apply the same CAC file uniformly across all images to ensure consistency.
9. Select the root area to be analysed and enter the sample identification details.
10. Following the analyses, the graphical display will present color assessments derived from the uploaded CAC file.
11. Export raw area measurements corresponding to each color class as text files (see **Note 5**).
12. Calculate the percent disease severity index (DSI) following Chiang et al. (2017) using the formula:

$$\text{Disease severity index (\%)} = \frac{(9A+8B+7C+6D+5E+4F+3G+2H+1I+0J)}{9 \times N} \times 100$$

13. Here, A–J represent areas corresponding to color classes 9–0, respectively, and *N* denotes the total root area analyzed.

14. Calculate the percent area infected for individual color classes as:

$$\text{Percent area infected} = \frac{\text{Area of specific color class}}{\text{Total root area}} \times 100$$

15. Assign disease severity ratings based on DSI values using a 0-5 scale (Table 2) (see **Note 6**).

Disease scoring using ImageJ

1. Import images into ImageJ software and calibrate using a known scale.
2. For camera and root scanner images, crop images to retain only root tissue using the rectangle selection tool.
3. Skip additional curation for microscopic images.
4. Adjust color threshold settings to select the entire root area.
5. Measure total root area using the “Analyze Particles” function.
6. Modify threshold settings to isolate necrotic regions and repeat particle analysis to measure necrotic area.
7. Convert area measurements from micrometres to centimetres.
8. Calculate percent necrotic area as:

$$\text{Percent necrotic area} = \frac{\text{Necrotic area}}{\text{Total root area}} \times 100$$

4. Notes

1. The seeds can be procured from ICRISAT (International Crops Research Institute for the Semi-Arid Tropics), Patancheru, Hyderabad, Telangana.
2. The soil-borne necrotrophic fungus was isolated from the roots of an infected plant collected from a field in Kamalapuram, Andhra Pradesh. The pure culture was obtained using the hyphal tip method. The same can be procured from ITCC.
3. Adjust the parameters (saturation, tolerance, hue, etc.) for each color class to include class-specific pixels to fall into the respective classes.

4. Enter the DPI (dots per inch) details for camera and microscope images for calibration.
5. Data files are in ASCII (text) format & easily readable in Excel.
6. For control images the DSI was falling under 0-15%, in the software the settings could not be changed for an immune genotype so that the DSI will be from 1%.

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